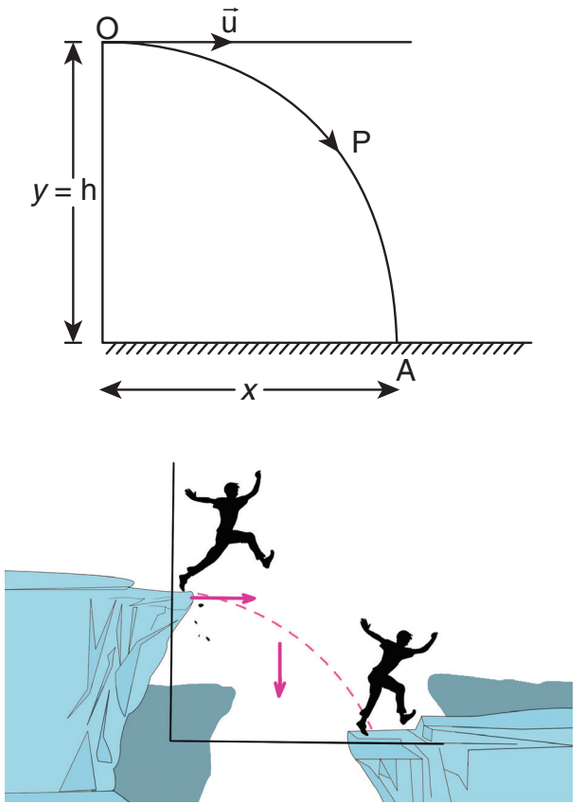


Figure 2.39 Horizontal Projection.

velocity u, and a vertical downward distance because of constant acceleration due to gravity g. Thus, under the combined effect the ball moves along the path OPA. The motion is in a 2-dimensional plane. Let the ball take time t to reach the ground at point A, Then the horizontal distance travelled by the ball is x (t) = x, and the vertical distance travelled is y (t) = y

We can apply the kinematic equations along the x direction and y direction separately. Since this is two-dimensional motion, the velocity will have both horizontal component u_x and vertical component u_y.

Motion along horizontal direction

The particle has zero acceleration along x direction. So, the initial velocity u_x remains constant throughout the motion.

The distance traveled by the projectile at a time t is given by the equation $x = u_x t + \frac{1}{2} a t^2$. Since a = 0 along x direction, we have

$$x = u_x t \qquad (2.23)$$

Motion along downward direction

Here u_y = 0 (initial velocity has no downward component), a = g (we choose the +ve y-axis in downward direction), and distance y at time t

$\therefore$ From equation, $y = u_y t + \frac{1}{2} a t^2$, we get

$$y = \frac{1}{2} g t^2 \qquad (2.24)$$

Substituting the value of t from equation (2.23) in equation (2.24) we have

$$y = \frac{1}{2} g \frac{x^2}{u_x^2} = \left(\frac{g}{2u_x^2} \right) x^2$$

$$y = Kx^2 \qquad (2.25)$$

where $K = \frac{g}{2u_x^2}$ is constant

Equation (2.25) is the equation of a parabola. Thus, the path followed by the projectile is a parabola (curve OPA in the Figure 2.39).

(1) Time of Flight: *The time taken for the projectile to complete its trajectory or time taken by the projectile to hit the ground is called time of flight.*

Consider the example of a tower and projectile. Let h be the height of a tower. Let T be the time taken by the projectile to hit the ground, after being thrown horizontally from the tower.

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